

$$m \vec{a}_{\Sigma, \text{app}} = \vec{F} - m 2 (\vec{\omega} \times \vec{v}_{\Sigma, \text{app}})$$

$$\vec{F} = \vec{F}_g = -mg \hat{e}_z$$

$$\vec{a}_{\text{app}} = -g \hat{e}_z - 2 (\vec{\omega} \times \vec{v}_{\text{app}})$$

$$\vec{\omega} \cdot \hat{e}_x = 0$$

$$\vec{\omega} \cdot \hat{e}_y = 0$$

$$\begin{aligned} \vec{\omega} \cdot \hat{e}_z &= \omega \cos\left(\frac{\pi}{2} - \theta\right) \\ &= \omega \sin(\theta) \end{aligned}$$

$$\begin{aligned} \vec{\omega} &= (0, \omega \sin\left(\frac{\pi}{2} - \theta\right), \omega \cos\left(\frac{\pi}{2} - \theta\right)) \\ &= (0, \omega \cos(\theta), \omega \sin(\theta)) \end{aligned}$$



